

Rishit Dutta

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Education

Indian Institute of Technology, Kanpur

Jul. 2024 – Present

BTech. in Electrical Engineering

- CGPA: 8.7/10.0
- **Relevant Coursework:** Fundamentals of Data Science & Machine Learning, Large Language Models, Probability & Statistics, Linear Algebra, Ordinary and Partial Differential Equations, Multivariate Calculus

Publications & Preprints

FrameNav: Deciding, Remembering and Monitoring in the Camera Frame for Zero-Shot VLN-CE

*WACV 2027
(Under Review)*

Rishit Dutta*, Toshith*, Tushar Sandhan

Do Adaptive Image Tokenizers Spend Tokens Where Vision Needs Them?

*NewInML, NeurIPS 2026
(Under Review)*

Rishit Dutta

Research Experiences

Research Assistant (Zero-Shot Vision-Language Navigation)

Aug. 2025 – Present

Advisor: Prof. Tushar Sandhan, IIT Kanpur

- Proposed FrameNav, a zero-shot VLN-CE framework unifying planning, memory, and localization in image space.
- Developed DQP for optical-flow localization (9.1 cm median error over 610 m) and topological memory with spectral graph-transformer retrieval, improving success from 28.6% to 57.1%.

Research Assistant (Sign Language Tokenization)

May. 2026 – Present

Advisor: Prof. Ashutosh Modi, IIT Kanpur

- Implemented a variable-length sign-video tokenizer using discrete latent quantization methods (VQ-VAE, RQ-VAE, FSQ), and evaluated pose-reconstruction and contrastive sign-text objectives.
- Built How2Sign/YouTube-ASL preprocessing at scale (500K+ clips, 5.5M+ pose frames) across face, hand, body, and pose representations.

Research Assistant (Multimodal Representation Collapse)

April. 2026 – Present

Advisor: Prof. Rajesh Hegde, IIT Kanpur

- Evaluated EBR as a representation-collapse intervention on Audio-Visual MNIST, improving fusion similarity by 18.6% and numerical rank by 73%.
- Extended multimodal representation-collapse analysis to VLMs using representation-geometry diagnostics and persistent homology.

Research Assistant (Deep Learning for Source Apportionment and Forecasting)

Aug. 2025 – Jun. 2026

Advisor: Prof. Rajesh Hegde, IIT Kanpur

- Developed neural models for sector-wise emission source apportionment from EDGAR/ERA5 data, benchmarking against K-means and GMMs, achieving 0.012 KL Divergence.
- Designed a wind-directed LSTM-GAT for PM2.5/PM10 forecasting, modeling wind through graph edges and reducing RMSE by 36% vs. GCN.

Scholastic Achievements

JEE Mains

2024

Top 0.3% among ≈ 14.8 million candidates

JEE Advanced

2024

All India Rank 1020 among 180,200 candidates

Indian National Astronomy Olympiad (INAO)

2022

National Top 2% among 13,201 candidates

National Standard Examination in Physics (NSEP)

2024

Awarded State Topper for being among the Top 1%

Under Graduate Projects

2025 & 2026

Achieved 10/10 Grade in all Undergraduate Project (UGP) courses taken

Skills

- **Programming Languages:** C, C++, Python
- **Software/Libraries:** NumPy, Pandas, Scikit-learn, PyTorch, Matplotlib, Seaborn, OpenCV
- **Tools/Utilities:** Git/Github, L^AT_EX